

The empirical recurrence for OEIS A281718, recovered from its tail

Adrian Perez Fontelles
Independent researcher

8 September 2026

Abstract

OEIS A281718 records “Empirical recurrence of order 72” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 274$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 74$. Its last coefficient is $c_{72} = -1 \neq 0$, so no further tail satisfies anything shorter; any order-72 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A281718 is “Number of $4 \times n$ 0..1 arrays with no element unequal to a strict majority of its horizontal, diagonal and antidiagonal neighbors and with new values introduced in order 0 sequentially upwards.”. It has offset 1 and begins

8, 6, 14, 97, 294, 1410, 5781, 23798, ...

The entry states:

Empirical recurrence of order 72 (see link above)

As of the “Last modified” line on the live entry (Jan 28 2017 13:48:31, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 274$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 274$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 72: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 72.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 548$ exact terms from $n = 2$ on, it returns order 72 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{72} c_i a(n-i),$$

c_1	5	c_{19}	−28516836	c_{37}	1491835997	c_{55}	14484570
c_2	27	c_{20}	−156762042	c_{38}	1296643709	c_{56}	−14040705
c_3	−117	c_{21}	78716581	c_{39}	−1055088405	c_{57}	−5106544
c_4	−498	c_{22}	330866772	c_{40}	−1008309268	c_{58}	4609382
c_5	1426	c_{23}	−191551733	c_{41}	591399202	c_{59}	1476162
c_6	6166	c_{24}	−605372549	c_{42}	755434060	c_{60}	−1250493
c_7	−10999	c_{25}	403634208	c_{43}	−221265782	c_{61}	−346343
c_8	−53195	c_{26}	964506675	c_{44}	−549260269	c_{62}	274789
c_9	59738	c_{27}	−731840268	c_{45}	−960398	c_{63}	64599
c_{10}	338792	c_{28}	−1345645521	c_{46}	383355727	c_{64}	−47621
c_{11}	−251260	c_{29}	1141073860	c_{47}	88727883	c_{65}	−9241
c_{12}	−1677363	c_{30}	1654867415	c_{48}	−250536777	c_{66}	6261
c_{13}	897487	c_{31}	−1531880047	c_{49}	−92436641	c_{67}	953
c_{14}	6728173	c_{32}	−1807687458	c_{50}	148952885	c_{68}	−587
c_{15}	−2961367	c_{33}	1771673007	c_{51}	63442503	c_{69}	−63
c_{16}	−22527022	c_{34}	1770700182	c_{52}	−78514898	c_{70}	35
c_{17}	9424065	c_{35}	−1760987317	c_{53}	−33698893	c_{71}	2
c_{18}	64127029	c_{36}	−1575669316	c_{54}	35937505	c_{72}	−1

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 74$, and it is the recurrence OEIS A281718 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{72} - c_1 x^{72-1} - \dots - c_{72}$. Its constant term is $-c_{72} = 1$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$

vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 72 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 72, so $\deg q = 72$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 72, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 23 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 72, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -1 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-72 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A281718.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)
- [5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.